



Volatility Surface

Keith Law

Concept of Implied Volatility

- The story starts with Black-Scholes

$$C(S, t) = N(d_1)S - N(d_2)Ke^{-r(T-t)}$$

$$d_1 = \frac{1}{\sigma\sqrt{T-t}} \left[\ln\left(\frac{S}{K}\right) + \left(r + \frac{\sigma^2}{2}\right)(T-t) \right]$$

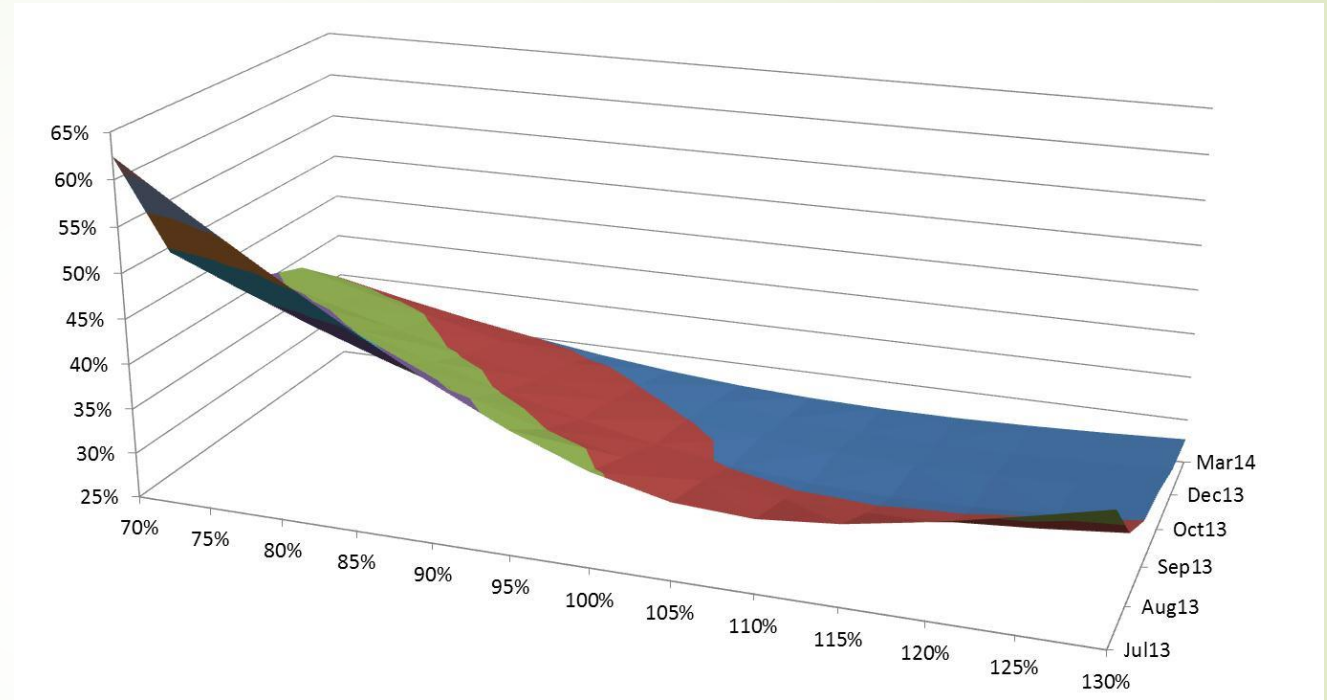
$$d_2 = \frac{1}{\sigma\sqrt{T-t}} \left[\ln\left(\frac{S}{K}\right) + \left(r - \frac{\sigma^2}{2}\right)(T-t) \right]$$
$$= d_1 - \sigma\sqrt{T-t}$$

$$P(S, t) = Ke^{-r(T-t)} - S + C(S, t)$$
$$= N(-d_2)Ke^{-r(T-t)} - N(-d_1)S$$

- Option pricing takes in 6 parameters: Spot, Strike, Interest rate, Dividend, Maturity, **Volatility**
- B&S** takes in volatility independent of strike and maturity, i.e. a constant volatility surface
- However, we note that the option prices with lower strikes and far upper strikes do NOT follow the B&S prices, and are generally **higher**
- Therefore, we tweak the playing field: What volatility when plugged into the B&S formula give the prices we observed in the market, i.e. **Implied Volatility**
- Essentially, using the wrong input into the wrong formula to give the right output!
- The game is then modeling the volatility dynamics (skews / smiles) in a parameterization form

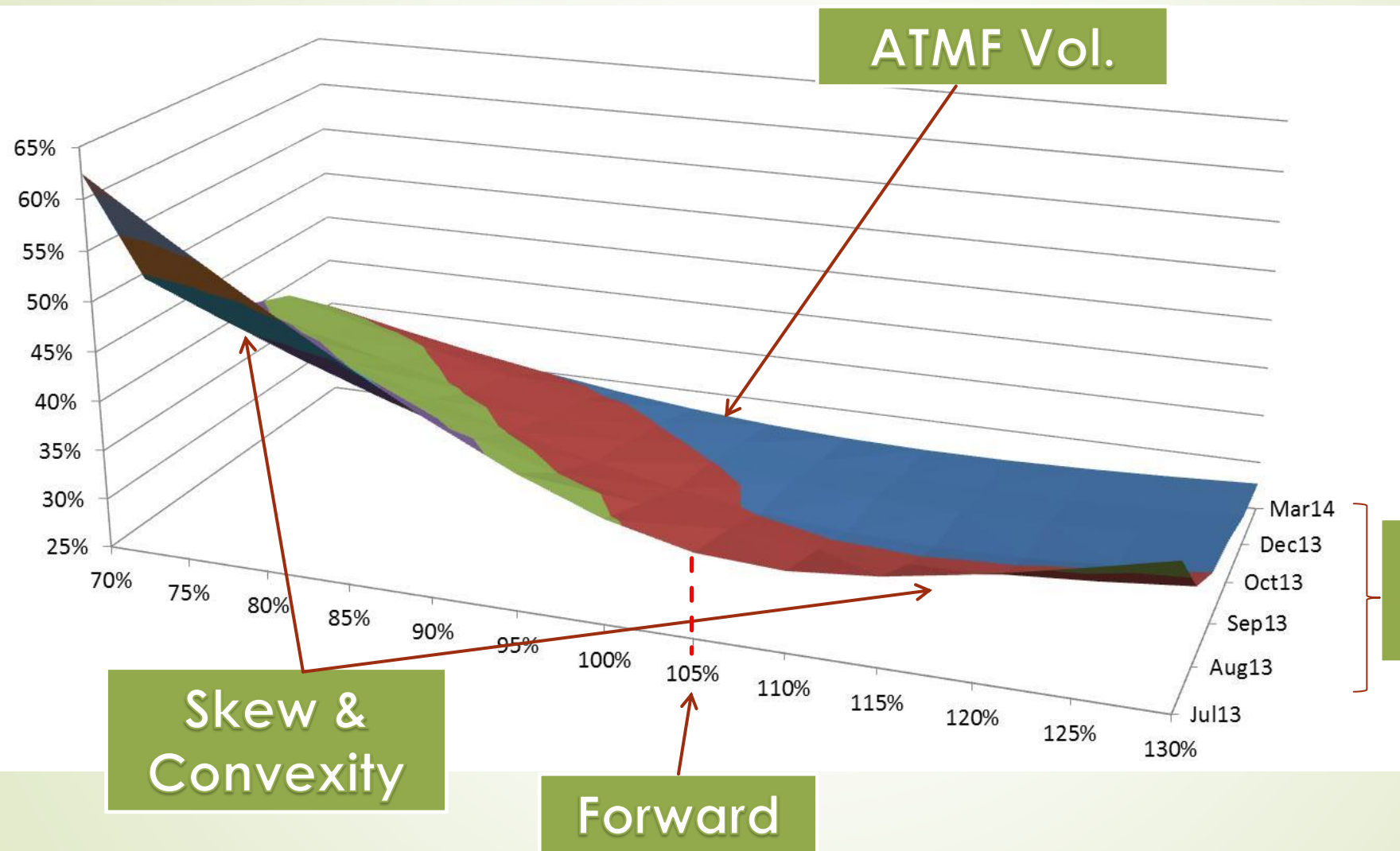
Implied Volatility Surface

- Plot **Implied Volatility** against:
 - Time to Maturity (y-axis)
 - Strikes / Moneyness (x-axis)
- Skew / Smile** usually more prevalent in the short-term
- Dynamics:** Change of spot => a shift of surface
- Vol surface parameterization
 - Fit the market prices well
 - Model the dynamics well



	70.00%	75.00%	80.00%	85.00%	90.00%	95.00%	100.00%	105.00%	110.00%	115.00%	120.00%	125.00%	130.00%
Jul13	62.49%	57.14%	52.04%	47.21%	42.69%	38.66%	35.41%	33.42%	32.96%	33.75%	35.28%	37.17%	39.20%
Aug13	49.42%	46.06%	42.93%	40.02%	37.38%	35.07%	33.19%	31.85%	31.11%	30.95%	31.26%	31.91%	32.77%
Sep13	45.15%	42.20%	39.45%	36.91%	34.60%	32.58%	30.93%	29.71%	28.98%	28.74%	28.90%	29.37%	30.03%
Oct13	40.78%	38.47%	36.34%	34.41%	32.70%	31.24%	30.08%	29.23%	28.72%	28.52%	28.60%	28.88%	29.33%
Dec13	38.87%	36.67%	34.65%	32.81%	31.18%	29.80%	28.69%	27.88%	27.38%	27.19%	27.25%	27.52%	27.94%
Mar14	34.63%	32.91%	31.36%	30.01%	28.86%	27.95%	27.28%	26.87%	26.69%	26.73%	26.94%	27.28%	27.73%

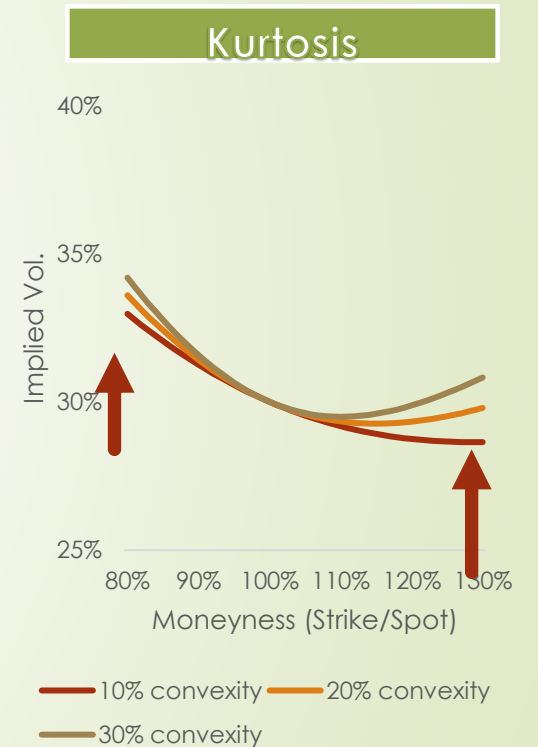
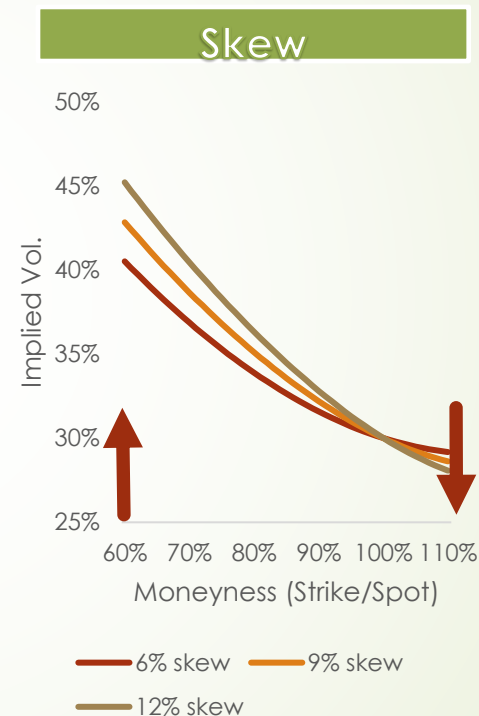
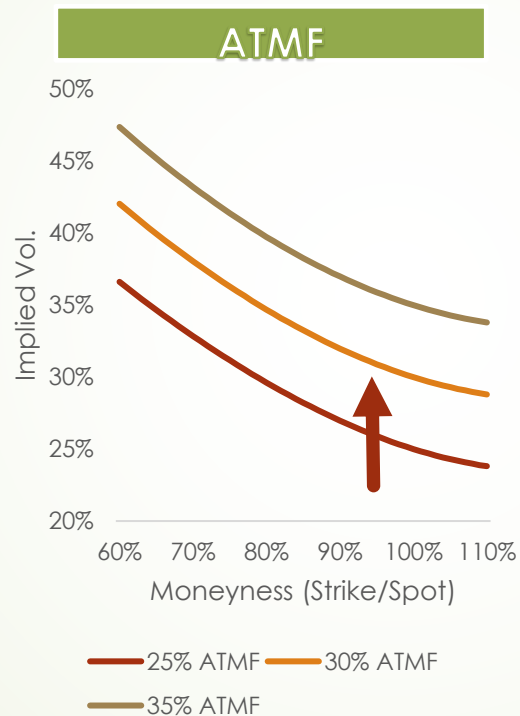
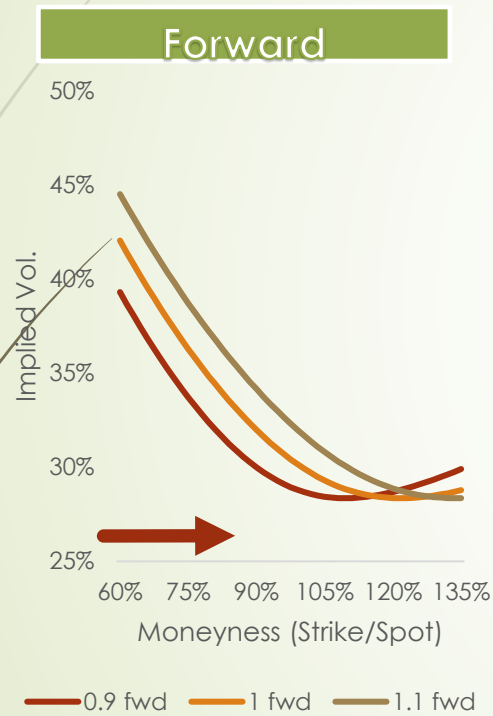
Parameterization of Volatility Surface



Landscape of Volatility Surface Modelling

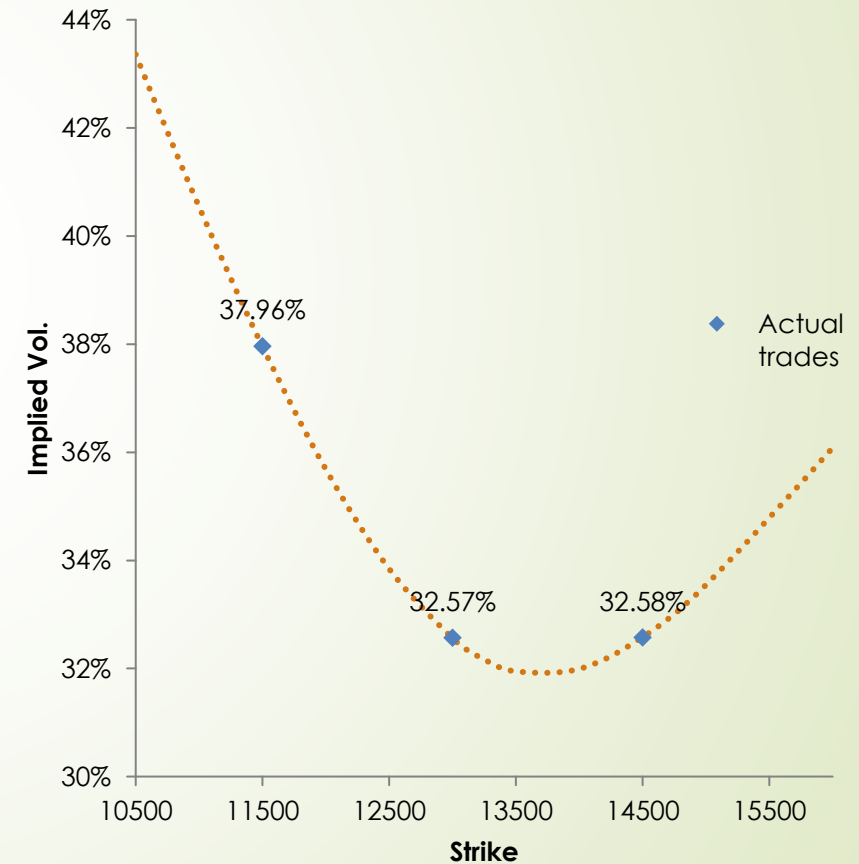
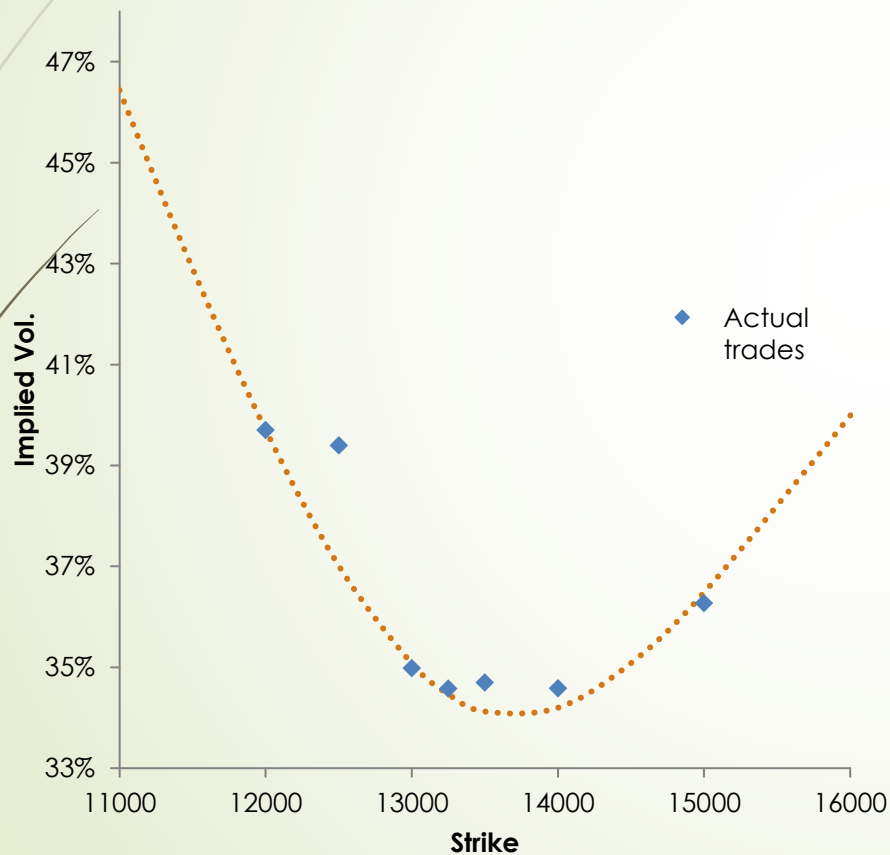
- The functional forms of vol surface fitting is the job of Quantitative Research at different banks
- There is NO unique / consensus function
- However, the functional form should take the following into accounts
 - Where the Forward? ATMF vol
 - Steepness of skew
 - Celerity of the wings
 - Dynamics of the surface (how skew change when there is a change in spot/ forward)
- The functional form usually takes in some trigonometric functions

Visualization of parameters' effect



Calibration of Volatility Surface

- Assume **skew** and **convexity** to fit a continuous quadratic function connecting discrete points



An example

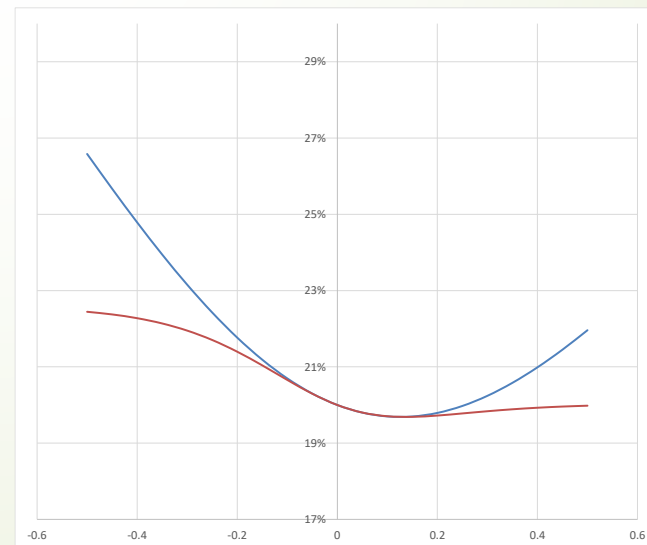
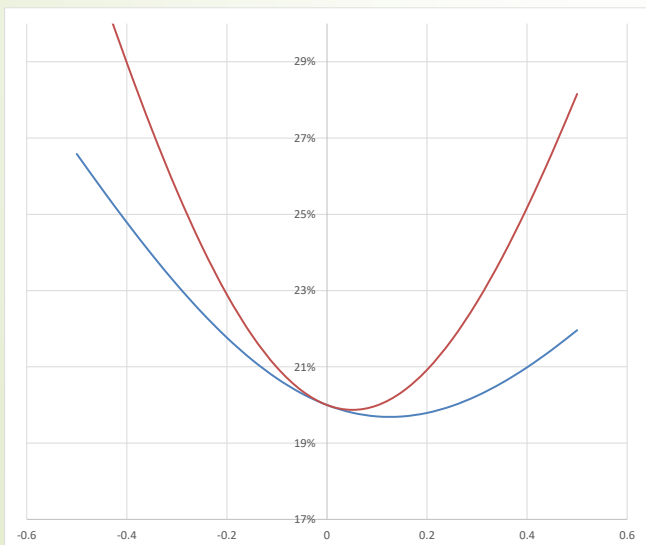
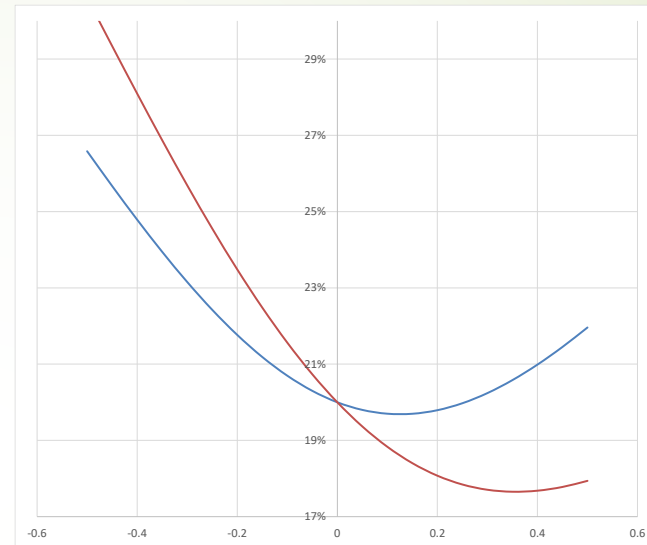
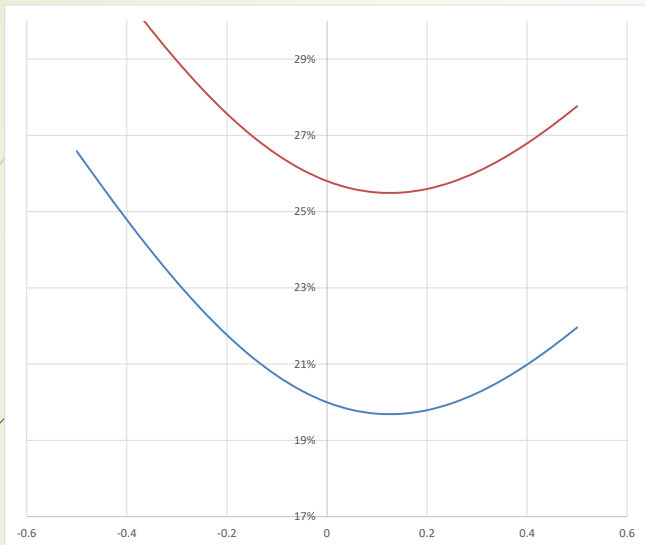
- $\sigma(x)^2 = \sigma_{atm}^2 + \delta \left(\frac{\tanh(\kappa x)}{\kappa} \right) + \frac{\gamma}{2} \left(\frac{\tanh(\kappa x)}{\kappa} \right)^2$

- Where x is the moneyness, $x = \ln \left(\frac{K}{S} \right)$, δ is the skew, γ is the convexity, κ

controls the 'wings'

- Instead of tanh, you may use sigmoid function, the key is to die down the 'tail' on both sides

Changing vol, skew, convexity, celerity





Local Volatility vs Stochastic Volatility

Keith Law

Local Volatility (Dupire Equation)

- $\frac{dS}{S} = \mu(t)dt + \sigma(S, t)dZ$
- If we express the option price as a function of the forward price

$$F_T = S_0 \exp\left\{\int_0^T \mu(t)dt\right\}$$

$$\frac{\partial C}{\partial T} = \frac{\sigma^2 K^2}{2} \frac{\partial^2 C}{\partial K^2}$$

Inverting gives (see Gatheral's paper for details)

$$\sigma^2(K, T, S_0) = \frac{\frac{\partial C}{\partial T}}{\frac{1}{2} K^2 \frac{\partial^2 C}{\partial K^2}}$$

The right hand side of equation can be computed from known European option prices

- So given a complete set of European option prices for all strikes and expirations, local volatilities are given uniquely
- We can view the above as a definition of the local volatility function regardless of what kind of process (stochastic process for example) actually governs the evolution of volatility

Local Volatility vs Black-Scholes Implied vol

- Market prices of options are quoted in terms of Black-Scholes implied volatility $\sigma_{BS}(K, T; S_0)$. We may write $C(S_0, K, T) = C_{BS}(S_0, K, \sigma_{BS}(S_0, K, T), T)$
- Define $w(S_0, K, T) \equiv \sigma_{BS}^2(S_0, K, T)T$, which is the 'total volatility' and log-strike $y = \ln\left(\frac{K}{F_T}\right)$, the BS formula for call becomes

$$C_{BS}(F_T, y, w) = F_T \{N(d_1) - e^y N(d_2)\} = F_T \left\{ N\left(-\frac{y}{\sqrt{w}} + \frac{\sqrt{w}}{2}\right) - e^y N\left(-\frac{y}{\sqrt{w}} - \frac{\sqrt{w}}{2}\right) \right\}$$

- Dupire becomes

$$\frac{\partial C}{\partial T} = \frac{v_L}{2} \left\{ \frac{\partial^2 C}{\partial y^2} - \frac{\partial C}{\partial y} \right\} + \mu(T)C \quad (*)$$

where $v_L = \sigma^2(S_0, K, T)$

- Taking derivatives of the Black-Scholes

$$\begin{aligned} \frac{\partial^2 C_{BS}}{\partial w^2} &= \left(-\frac{1}{8} - \frac{1}{2w} + \frac{y^2}{2w^2} \right) \frac{\partial C_{BS}}{\partial w} \\ \frac{\partial^2 C_{BS}}{\partial y \partial w} &= \left(\frac{1}{2} - \frac{y}{w} \right) \frac{\partial C_{BS}}{\partial w} \\ \frac{\partial^2 C_{BS}}{\partial y^2} - \frac{\partial C_{BS}}{\partial y} &= 2 \frac{\partial C_{BS}}{\partial w} \end{aligned}$$

Local Volatility vs Black-Scholes Implied vol

- ▶ We may transform in terms of implied variance by making the substitutions

$$\frac{\partial C}{\partial y} = \frac{\partial C_{BS}}{\partial y} + \frac{\partial C_{BS}}{\partial w} \frac{\partial w}{\partial y}$$

$$\frac{\partial^2 C}{\partial y^2} = \frac{\partial^2 C_{BS}}{\partial y^2} + 2 \frac{\partial^2 C_{BS}}{\partial y \partial w} \frac{\partial w}{\partial y} + \frac{\partial^2 C_{BS}}{\partial w^2} \left(\frac{\partial w}{\partial y} \right)^2 + \frac{\partial C_{BS}}{\partial w} \frac{\partial^2 w}{\partial y^2}$$

$$\frac{\partial C}{\partial T} = \frac{\partial C_{BS}}{\partial T} + \frac{\partial C_{BS}}{\partial w} \frac{\partial w}{\partial T} = \underbrace{\frac{\partial C_{BS}}{\partial w} \frac{\partial w}{\partial T}}_{\frac{v_L}{2} \left\{ \frac{\partial^2 C}{\partial y^2} - \frac{\partial C}{\partial y} \right\} \text{ compared to (*)}} + \mu(T) C_{BS}$$

$$\frac{v_L}{2} \left\{ \frac{\partial^2 C}{\partial y^2} - \frac{\partial C}{\partial y} \right\} \text{ compared to (*)}$$

Local Volatility vs Black-Scholes Implied vol

- Where the last equality follows from the fact that the only explicit dependence of the option price on T is through the forward price $F_T = S_0 \exp\{\int_0^T \mu(t) dt\}$

$$\begin{aligned} & \frac{\partial C_{BS}}{\partial w} \frac{\partial w}{\partial T} \\ &= \frac{v_L}{2} \left\{ -\frac{\partial C_{BS}}{\partial y} + \frac{\partial^2 C_{BS}}{\partial y^2} - \frac{\partial C_{BS}}{\partial w} \frac{\partial w}{\partial y} + 2 \frac{\partial^2 C_{BS}}{\partial y \partial w} \frac{\partial w}{\partial y} + \frac{\partial^2 C_{BS}}{\partial w^2} \left(\frac{\partial w}{\partial y} \right)^2 + \frac{\partial C_{BS}}{\partial w} \frac{\partial^2 w}{\partial y^2} \right\} \\ &= \frac{v_L}{2} \frac{\partial C_{BS}}{\partial w} \left\{ 2 - \frac{\partial w}{\partial y} + 2 \left(\frac{1}{2} - \frac{y}{w} \right) \frac{\partial w}{\partial y} + \left(-\frac{1}{8} - \frac{1}{2w} + \frac{y^2}{2w^2} \right) \left(\frac{\partial w}{\partial y} \right)^2 + \frac{\partial^2 w}{\partial y^2} \right\} \end{aligned}$$

$$\frac{\partial w}{\partial T} = v_L \left\{ 1 - \frac{y}{w} \frac{\partial w}{\partial y} + \frac{1}{4} \left(-\frac{1}{4} - \frac{1}{w} + \frac{y^2}{w^2} \right) \left(\frac{\partial w}{\partial y} \right)^2 + \frac{1}{2} \frac{\partial^2 w}{\partial y^2} \right\}$$

- Special case: No Skew: $\frac{\partial w}{\partial y} = 0$, then $v_L = \frac{\partial w}{\partial T}$, i.e. $w(T) = \int_0^T v_L(t) dt$

Local variance as a conditional expectation of instantaneous variance

- In terms of forward, $dF_{t,T} = \sqrt{v_t}F_{t,T}dZ$, note that $dF_{T,T} = dS_T$
- The undiscounted value of a European option with strike K expiring at time T is given by

$$C(S_0, K, T) = E[(S_T - K)^+]$$

- Differentiating once with respect to K gives

$$\frac{\partial C}{\partial K} = -E[\theta(S_T - K)]$$

Where $\theta(\cdot)$ is the Heaviside function. Differentiating again with respect to K gives

$$\frac{\partial^2 C}{\partial K^2} = E[\delta(S_T - K)]$$

$\delta(\cdot)$ is the Dirac-delta function

A formal application of Ito's lemma to the terminal payoff of the option (and using $dF_{T,T} = dS_T$) gives the identity

$$d(S_T - K)^+ = \theta(S_T - K)dS_T + \frac{1}{2}v_T S_T^2 \delta(S_T - K)dT$$

Local variance as a conditional expectation of instantaneous variance

- ▶ Taking conditional expectations of each side and using the fact that $F_{t,T}$ is a martingale, we get

$$dC = dE[(S_T - K)^+] = \frac{1}{2}E[v_T S_T^2 \delta(S_T - K)]dT$$

Also we can write

$$\frac{1}{2}E[v_T S_T^2 \delta(S_T - K)] = E[v_T | S_T = K] \frac{1}{2}K^2[\delta(S_T - K)] = E[v_T | S_T = K] \frac{1}{2}K^2 \frac{\partial^2 C}{\partial K^2}$$

Putting this together, we get,

$$\frac{\partial C}{\partial T} = E[v_T | S_T = K] \frac{1}{2}K^2 \frac{\partial^2 C}{\partial K^2}$$

We see $\sigma^2(K, T, S_0) = E[v_T | S_T = K]$